

JOROINEN VARKAUS, Finland

WMO: 027480

Lat: 62.174N

Lon: 27.860E

Elev: 286

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 99-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.9	-13.1	-25.2	1.4	-19.8	-18.2	2.1	-13.3	17.5	24.1	15.9	24.0	1.7	320	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	79.1	64.8	75.6	63.3	73.2	61.7	67.3	75.0	65.4	72.8	63.5	70.7	7.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	92.0	70.5	62.6	85.8	68.6	60.6	79.9	67.0	31.9	74.6	30.3	72.9	29.0	70.6	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.4	13.0	DB	-21.7	83.8	8.6	3.3	-27.9	86.2	-33.0	88.1	-37.8	90.0	-44.0	92.4	(4)
(5)			WB	-22.5	70.2	8.3	3.5	-28.5	72.7	-33.3	74.8	-37.9	76.7	-43.9	79.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	17.1	16.1	24.4	36.9	48.7	56.6	61.1	59.2	50.3	39.5	31.5	23.6	(6)
(7)		DBStd	18.11	13.91	14.02	10.07	6.87	7.52	6.52	4.99	5.51	6.02	7.26	8.27	12.76	(7)
(8)		HDD50	5065	1020	948	794	395	116	13	0	5	70	331	554	819	(8)
(9)		HDD65	9579	1485	1368	1259	842	508	264	141	191	441	792	1004	1284	(9)
(10)		CDD50	1006	0	0	0	2	75	211	344	290	79	5	0	0	(10)
(11)		CDD65	45	0	0	0	0	2	12	20	11	0	0	0	0	(11)
(12)		CDH74	491	0	0	0	0	31	127	215	118	0	0	0	0	(12)
(13)		CDH80	67	0	0	0	0	2	15	33	17	0	0	0	0	(13)
(14)	Wind	WSAvg	5.8	5.8	5.7	6.1	5.9	6.2	5.9	5.1	4.8	5.5	5.9	6.4	6.6	(14)
(15)	Precipitation	PrecAvg	23.8	1.7	1.5	1.3	1.3	1.6	2.5	3.3	2.7	1.9	2.3	2.0	1.8	(15)
(16)		PrecMax	27.8	3.0	2.8	2.8	2.6	3.0	4.4	5.3	5.2	4.2	4.3	4.4	3.1	(16)
(17)		PrecMin	19.6	0.4	0.2	0.1	0.4	0.6	1.0	1.1	0.9	0.9	0.4	0.2	0.6	(17)
(18)		PrecStd	2.1	0.7	0.7	0.6	0.6	0.8	0.9	1.1	1.2	0.9	1.0	0.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	39.7	50.6	64.0	78.6	82.3	84.0	82.1	70.0	57.0	49.8	45.0	(19)
(20)			MCWB	37.8	37.0	40.8	52.0	60.9	65.6	67.2	68.4	59.9	52.9	47.6	43.3	(20)
(21)		2%	DB	37.1	37.0	43.9	57.8	71.5	78.2	79.6	77.1	66.2	53.8	46.7	41.9	(21)
(22)			MCWB	36.1	35.4	37.4	47.2	56.9	63.9	65.9	65.2	58.5	51.2	45.2	40.7	(22)
(23)		5%	DB	35.2	34.6	40.0	53.3	66.9	73.8	75.6	73.7	63.1	51.9	44.5	39.3	(23)
(24)			MCWB	34.5	33.7	35.6	43.7	54.3	61.7	63.6	62.7	57.2	49.7	43.4	38.1	(24)
(25)		10%	DB	33.4	32.8	37.3	48.6	63.0	69.8	73.1	71.3	60.7	50.1	42.2	37.3	(25)
(26)			MCWB	33.0	32.0	34.4	41.6	52.4	59.1	61.8	61.8	55.8	48.5	41.3	36.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	37.5	41.7	51.9	62.8	69.2	70.3	70.0	61.8	54.4	47.9	43.6	(27)
(28)			MCD B	38.0	39.0	48.0	61.8	74.5	77.4	77.8	78.6	66.8	56.5	49.0	44.6	(28)
(29)		2%	WB	35.6	35.5	38.1	48.4	59.7	65.9	67.3	66.7	59.9	52.1	45.2	40.5	(29)
(30)			MCD B	36.2	36.6	42.3	54.9	67.9	73.4	75.8	74.1	64.6	53.7	46.3	41.3	(30)
(31)		5%	WB	33.9	33.7	36.2	45.1	56.5	63.5	65.4	64.9	58.0	50.5	43.0	38.1	(31)
(32)			MCD B	34.4	34.4	39.0	51.3	64.6	71.6	73.0	71.9	62.6	51.8	43.9	38.9	(32)
(33)		10%	WB	32.5	32.1	34.5	42.2	53.5	60.8	63.6	62.9	56.4	48.4	40.9	36.0	(33)
(34)			MCD B	32.9	32.7	36.4	47.2	62.0	68.7	70.9	69.2	60.2	49.6	41.7	36.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.7	12.9	15.9	16.1	18.3	16.9	18.0	16.9	14.7	9.4	7.2	9.0	(35)
(36)			MCDBR	7.6	8.6	17.5	23.3	25.1	23.3	23.5	23.5	19.3	11.2	7.9	6.8	(36)
(37)			MCWBR	7.6	8.0	12.2	13.7	13.0	11.1	11.5	12.3	12.4	8.9	7.4	6.7	(37)
(38)		5% WB	MCD BR	7.5	8.1	14.4	20.7	20.8	19.7	20.5	19.6	16.9	9.1	7.6	6.6	(38)
(39)			MCWBR	7.6	7.8	10.7	13.0	11.9	10.5	10.9	11.0	11.5	7.8	7.2	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.239	0.289	0.316	0.354	0.351	0.348	0.372	0.356	0.333	0.317	0.259	N/A	(40)
(41)		taud		2.034	2.122	2.190	2.341	2.453	2.488	2.437	2.466	2.522	2.405	2.128	N/A	(41)
(42)		Ebn at Noon		135	207	248	264	276	279	267	262	248	203	127	N/A	(42)
(43)		Edh at Noon		11	22	30	32	31	30	31	28	22	17	10	N/A	(43)
(44)	All-Sky Solar Radiation	RadAvg		67	248	732	1167	1549	1673	1610	1236	700	279	73	32	(44)
(45)		RadStd		10	60	95	133	173	101	156	147	80	47	12	4	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (39 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	-315	N/A	N/A	N/A

Nomenclature: See separate page